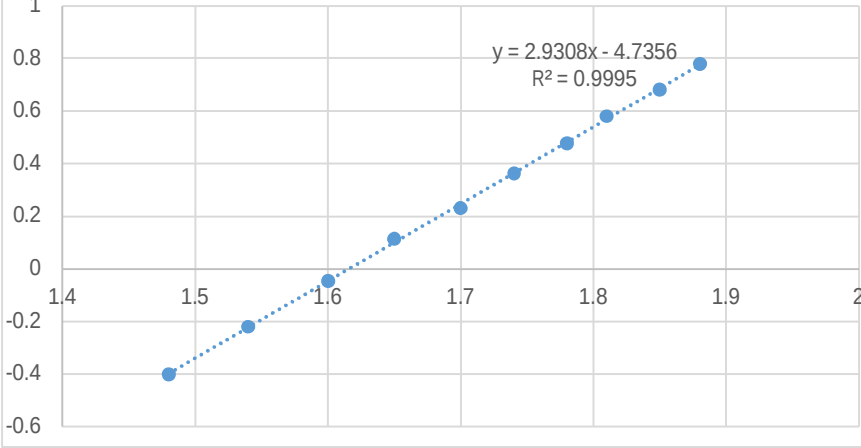
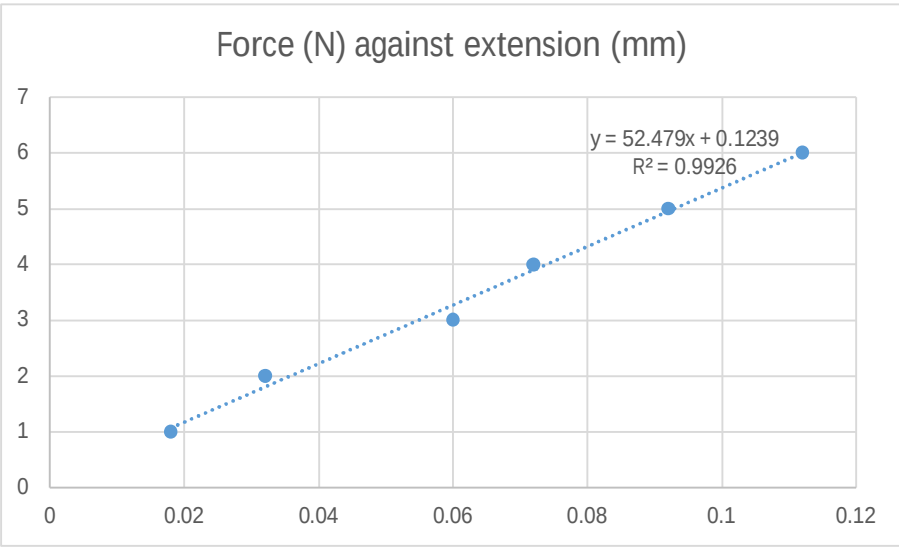


7	(a)	(i)	<table><tr><th>L / cm</th><th>d / cm</th><th>log (L /cm)</th><th>log (d /cm)</th></tr><tr><td>30</td><td>0.40</td><td>1.48</td><td>-0.40</td></tr><tr><td>35</td><td>0.60</td><td>1.54</td><td>-0.22</td></tr><tr><td>40</td><td>0.90</td><td>1.60</td><td>-0.046</td></tr><tr><td>45</td><td>1.30</td><td>1.65</td><td>0.114</td></tr><tr><td>50</td><td>1.70</td><td>1.70</td><td>0.230</td></tr><tr><td>55</td><td>2.30</td><td>1.74</td><td>0.362</td></tr><tr><td>60</td><td>3.00</td><td>1.78</td><td>0.477</td></tr><tr><td>65</td><td>3.80</td><td>1.81</td><td>0.580</td></tr><tr><td>70</td><td>4.80</td><td>1.85</td><td>0.681</td></tr><tr><td>75</td><td>6.00</td><td>1.88</td><td>0.778</td></tr></table>	L / cm	d / cm	log (L /cm)	log (d /cm)	30	0.40	1.48	-0.40	35	0.60	1.54	-0.22	40	0.90	1.60	-0.046	45	1.30	1.65	0.114	50	1.70	1.70	0.230	55	2.30	1.74	0.362	60	3.00	1.78	0.477	65	3.80	1.81	0.580	70	4.80	1.85	0.681	75	6.00	1.88	0.778	3
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		(ii)	log d vs log L  1 mark for correct labelling of axis with units 1 mark for plotting of all points 1 mark for best fit line	3																																												
		(iii)	Need to calculate gradient first and use substitution method to find y-intercept. Values of gradient in the range of 3.0 log k is the y-intercept which is -4.74. $k = 1.8 \times 10^{-5}$	1 1																																												

(b)	(i)	Force (N) against extension (mm)  1 mark for correct labelling of axis with units 1 mark for plotting of all points 1 mark for best fit line	3
	(ii)	Gradient = $\Delta y / \Delta x$ = 52.5 (within 10% tolerance)	1 1
	(iii)	$Y = (F / \Delta L) (L / A)$ = 52.5 [$1 \times 10^3 / (5.0 \times 10^{-5}) \times 10^6$] = 1.05 (1.1) $\times 10^3 \text{ N mm}^{-2}$	1 1
	(iv)	1. Steel / concrete/ stone/brick (reject glass, iron –rust, aluminum, plastic, metal-state the type?) 2. Since material is stiffer than wood, its Y should be greater than that of wood. The graph should have a gradient steeper than that of the wood. Correct graph of steeper gradient (take note of y-intercept of graph, should approach origin)	1 1 1
(c)	Moment about X = 0.20 (2.0) + 0.35 (5.0) + 0.80(1.5) = 3.35 (3.4) N m		1 1